Jerry (Jiarui) Zhang

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Education

University of Southern California **August 2022–Present**
Ph.D. in Computer Science, Advisor: Willie Neiswanger, GPA: 3.78/4.00 *Los Angeles, USA*

Tsinghua University **August 2018–June 2022**
Bachelor of Science in Electrical Engineering, GPA: 3.64/4.00 *Beijing, China*

Publications (* denotes equal contribution)

Google Scholar

- **J Zhang***, M Tao*, S Wang*, O Liu, X Ma, W Neiswanger. *'An Exam for Active Observers'*, arXiv 2026. [Paper] [Project]
- **J Zhang**, M Khayatkhoei, P Chhikara, F Ilievski. *'MLLMs Know Where to Look: Training-free Perception of Small Visual Details with Multimodal LLMs'*, ICLR 2025.
- **J Zhang**, O Liu, T Yu, J Hu, W Neiswanger. *'Euclid: Supercharging Multimodal LLMs with Synthetic High-Fidelity Visual Descriptions'*, In submission, 2025.
- **J Zhang***, J Hu*, M Khayatkhoei, F Ilievski, M Sun. *'Exploring Perceptual Limitations of Multimodal LLMs on Small Visual Objects'*, TMLR 2026.
- Y Jiang*, **J Zhang***, K Sun*, Z Sourati, K Ahrabian, K Ma, F Ilievski, J Pujara. *'MARVEL: Multidimensional Abstraction and Reasoning through Visual Evaluation and Learning'*, NeurIPS D&B Track 2024.
- **J Zhang**. *'Guided Profile Generation Improves Personalization with LLMs'*, EMNLP 2024.
- K Ahrabian, Z Sourati, K Sun, **J Zhang**, Y Jiang, F Morstatter, J Pujara. *'The curious case of nonverbal abstract reasoning with multi-modal large language models'*, COLM 2024.
- **J Zhang**, F Ilievski, K Ma, A Kollaa, J Francis, A Oltramari. *'A Study of Situational Reasoning for Traffic Understanding'*, KDD 2023.
- P Chhikara, **J Zhang**, F Ilievski, J Francis, K Ma. *'Knowledge-enhanced agents for interactive text games'*, K-CAP 2023. (Best Student Paper Award)
- **J Zhang**, F Ilievski, K Ma, J Francis, A Oltramari. *'A Study of Zero-shot Adaptation with Commonsense Knowledge'*, AKBC 2022.
- X Yi, **J Zhang**, W Li, X Wang, X Xie. *'Clickbait Detection via Contrastive Variational Modelling of Text and Label'*, IJCAI 2022.

Experience

University of Southern California **August 2022–Present**
Graduate Student Researcher *Los Angeles, USA*

- Developed training-free methods that use the attention and gradient maps of multimodal LLMs (MLLMs) to improve their perception of small visual details, raising the accuracy of two widely used MLLMs across seven visual question answering benchmarks without any training.
- Designed frontier multimodal benchmarks that expose capability gaps in leading MLLMs: ActiveVision for active observation across 17 tasks with human baselines, Geoperception for 2D geometric perception, and MARVEL for abstract visual reasoning across 770 puzzles.
- Analyzed how object quality, size, distractors, and location limit MLLM perception through controlled intervention studies; showed that attention maps locate the relevant region even when answers are incorrect; contributed new evaluation protocols for future MLLMs.
- Multimodal post-training: sweeping experiments across visual encoders, LLMs, training data composition, and curricula on high-fidelity synthetic data; delivered Euclid, a model strong at visual perception that also improves on multimodal reasoning benchmarks.

Google Cloud **August 2026–Present**
Research Intern *Sunnyvale, USA*

- Evaluate how well multimodal coding agents generate visual artifacts and communicate with target audiences.
- Explore vision-coding co-evolution, using visual feedback to improve how agents generate visual artifacts.

Google DeepMind **May 2025–August 2025**
Student Researcher *Los Angeles, USA*

- Built challenging, diagnostic benchmarks for multimodal language and reasoning models.

Amazon Alexa AI **May 2023–August 2023**
Applied Scientist Intern *Bellevue, USA*

- Improved the efficacy and efficiency of LLM-based personalization through step-wise self-optimization.